

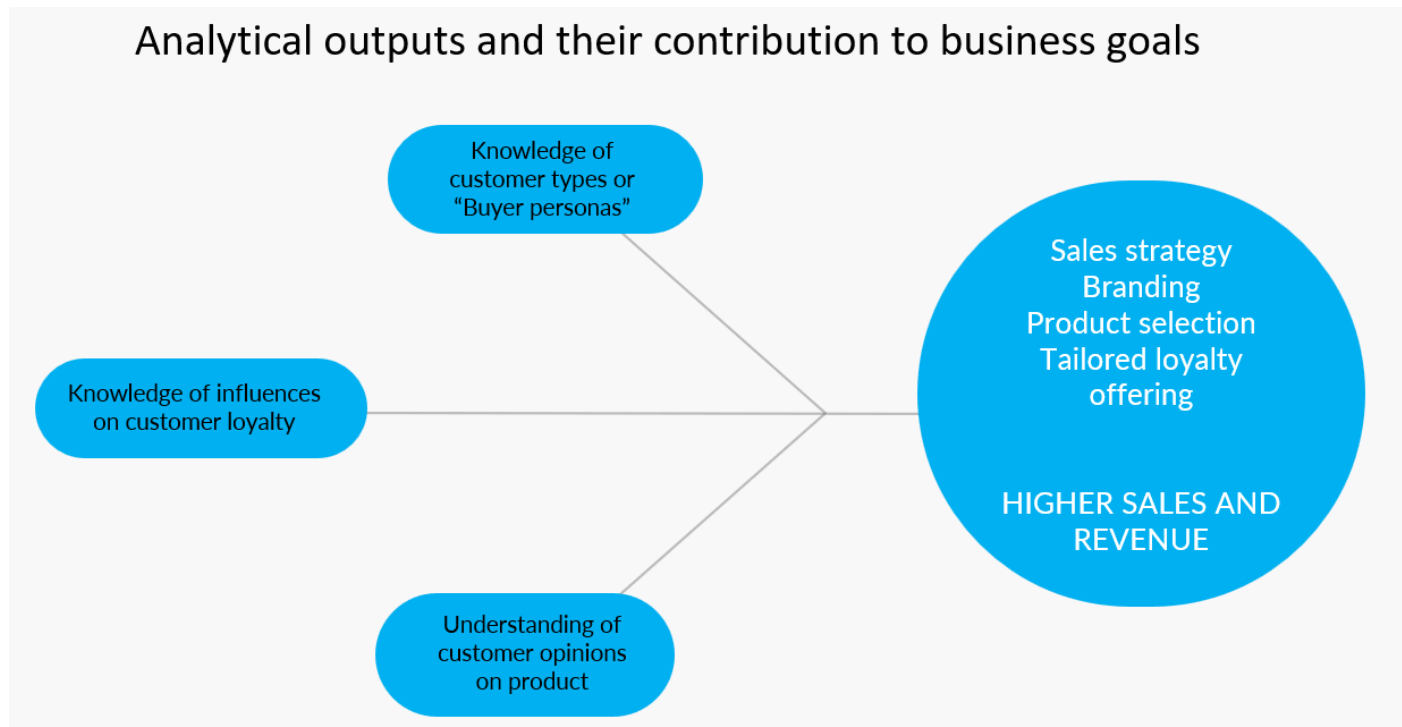
DA301 Advanced Analytics for Organisational Impact – Final Report

Introduction

Turtle Games has a sales growth objective. Strategies known to drive successful marketing and sales are:

- Developing customer insights via “buyer personas” (ref.1)
- Incentivising loyalty by integration with pricing (ref.3)
- Understanding review sentiment as a tool to improve future sales (ref.2)

The analysis aims to help Turtle Games understand its customers and their opinions and enable the implementation of these strategies. This can then inform sales efforts, branding, advertising choices and future product selection.



Analytical process and data exploration

Python and R were selected to conduct the analysis.

- Both widely used with libraries allowing efficient and powerful analyses.
- Code re-used to reduce effort and increase maintainability
- Python and R complement each other with strengths in different areas

In Python the data was imported using the Pandas library

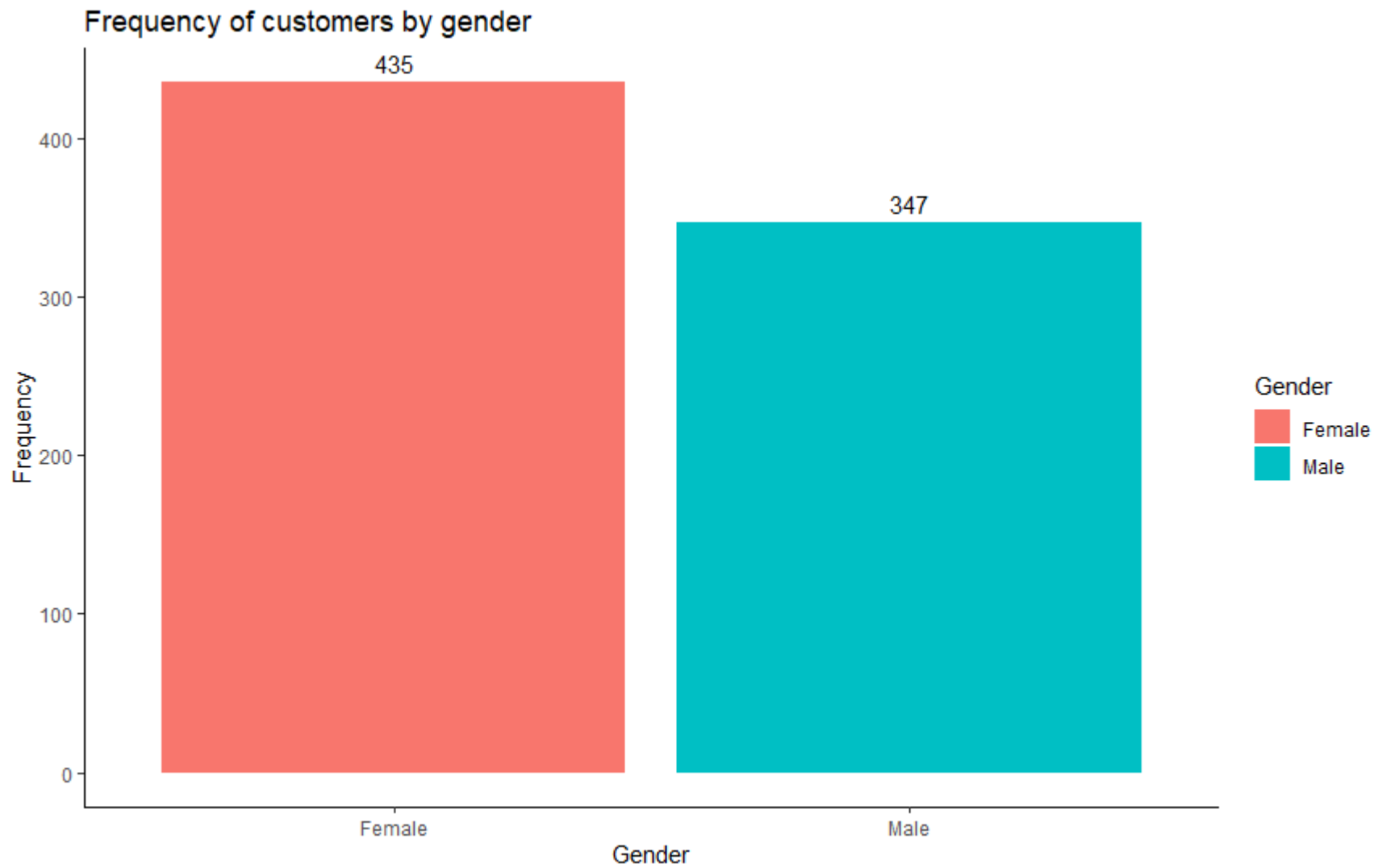
- Data checked for duplicates and missing values. None found.
- Redundant fields dropped.
- No outliers removed
- Data contained a row per review with a number of customer attributes, a product code, a review and a review summary.
- Data contains multiple reviews per customer, so two datasets were created
 - Cleaned data with all fields
 - Unique values of the customer attributes only

In R a similar process was followed but with different explorations to complement the Python work.

The following variables were explored in detail:

Gender

- 55.63% (435) of customers female and 44.37%(347) male.
- Small imbalance noted.

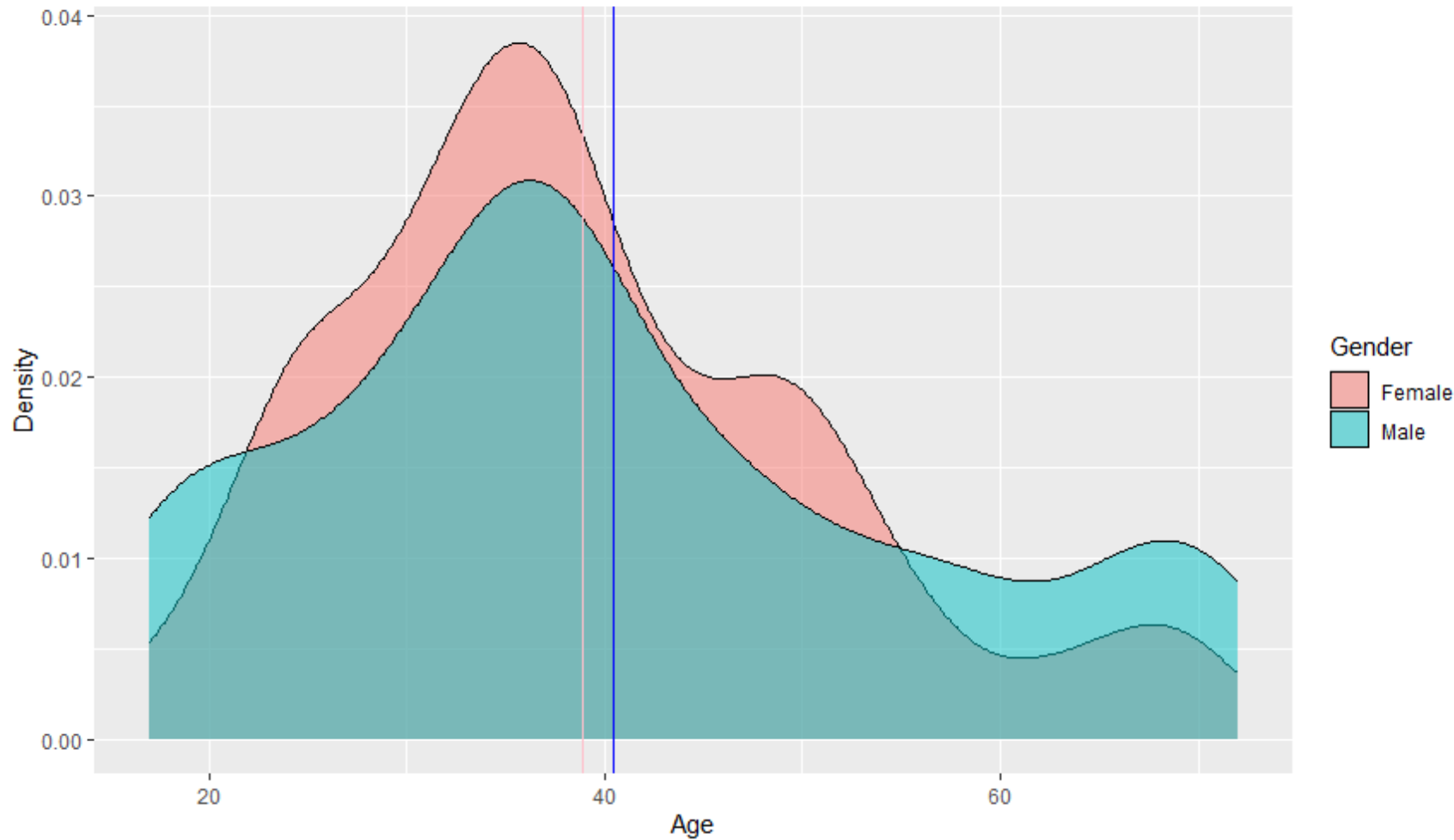


Age

- Minimum 17 and maximum 72, mean 39.5, median 38 and standard deviation 13.6.

Density plot of age by gender

Distribution skewed right for both genders: mean line shows female mean age slightly lower

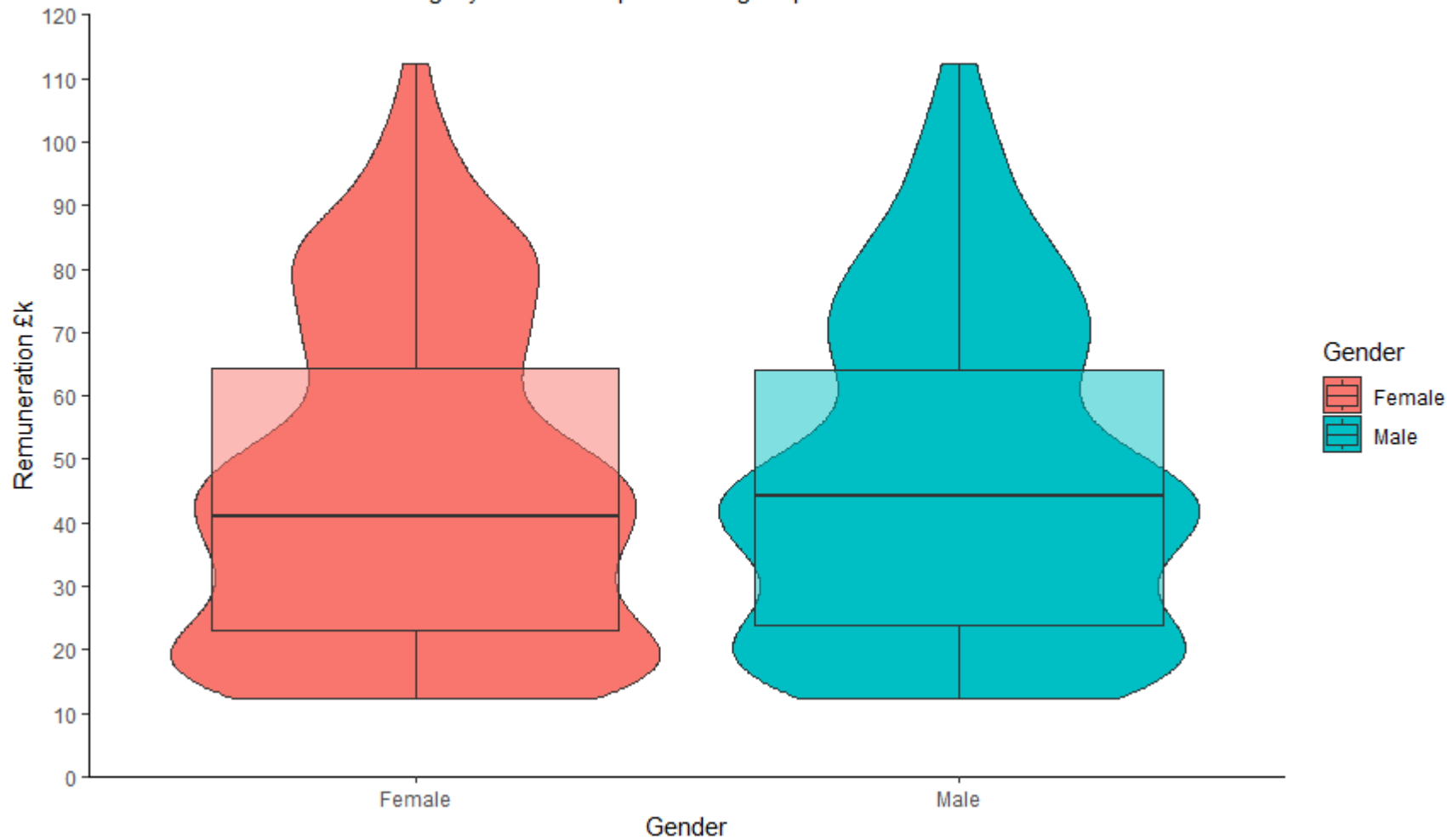


Remuneration

- Minimum 12.3, maximum 112.34, mean 46.06 and standard deviation: 25.23 (£K). Distribution is right-skewed.
- Mean well over the UK average of around 37K in 2025 (ref. 4).

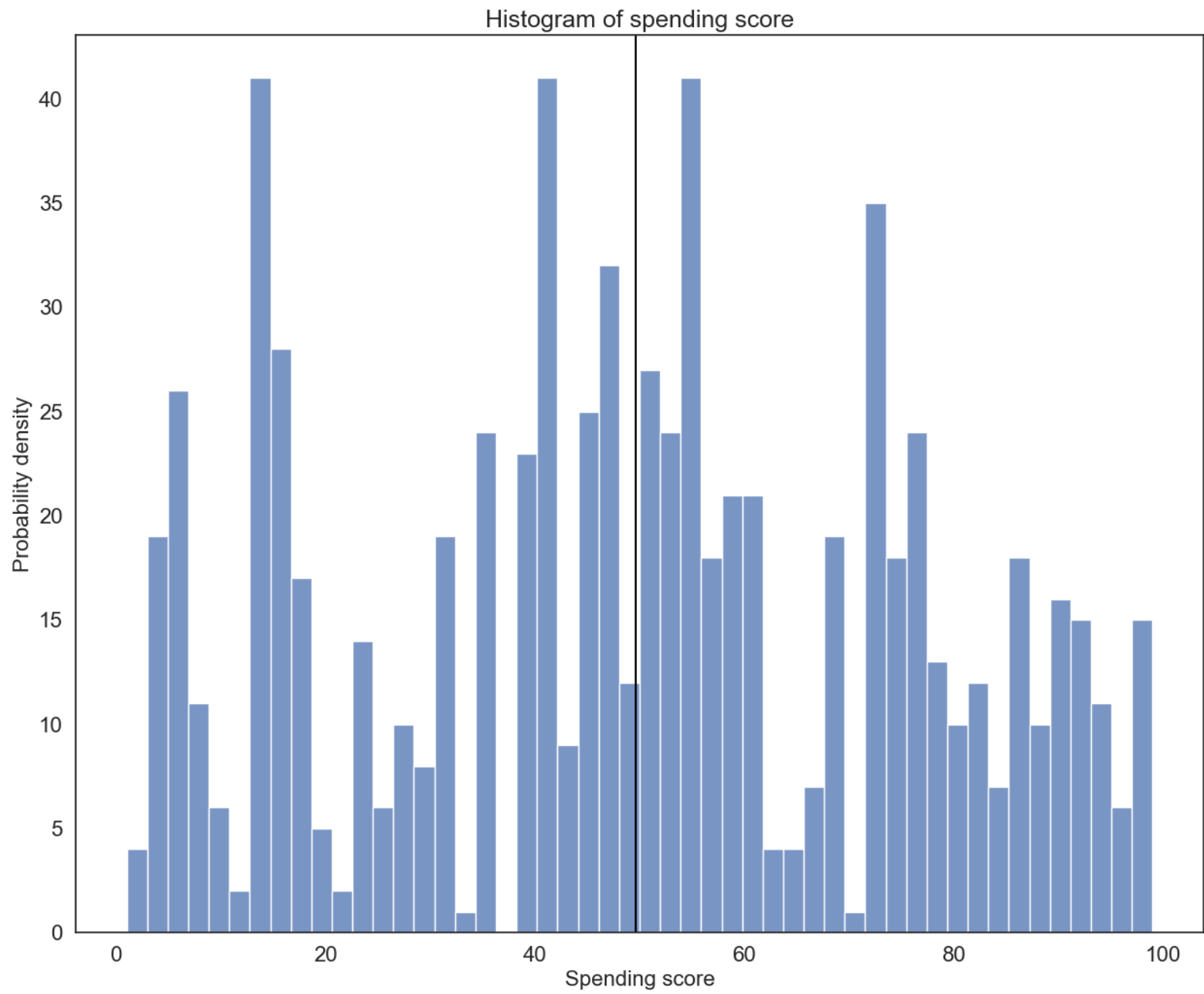
Box/Violin plot of remuneration by gender

Distributions similar but with slightly more lower paid and higher paid women



Spending score

- Minimum 1, maximum 99, mean 49.6 and standard deviation 26.58.
- Distribution symmetrical about the mean with “sub-peaks” at lower and higher values, suggesting clusters of low/high values.

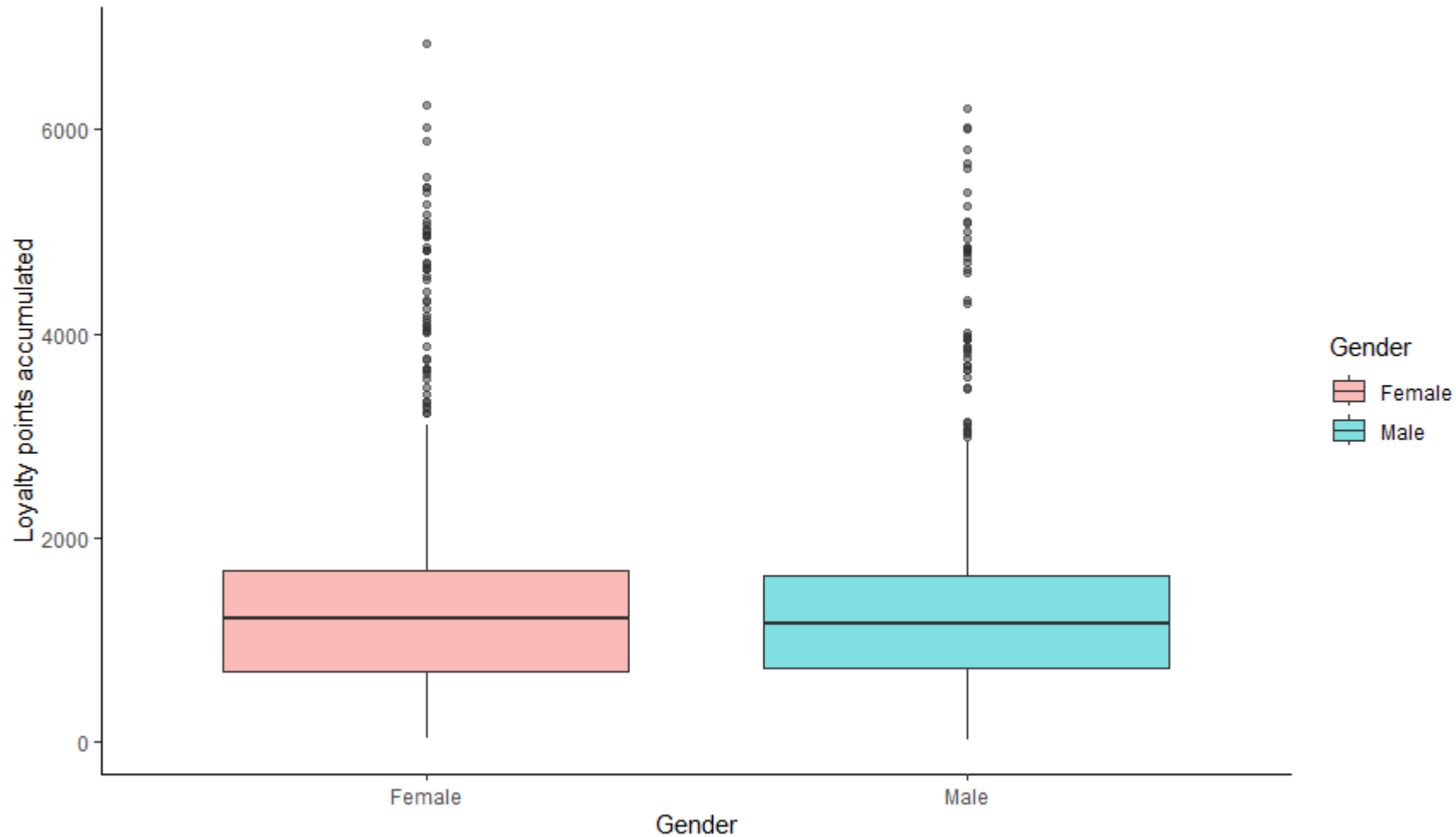


Loyalty points

- Right-skewed with minimum 25, maximum 6847, mean 1497.4 and standard deviation 1312.4.
- Some outliers with very high values > 1.5 IQR : not removed as they may represent important customers.

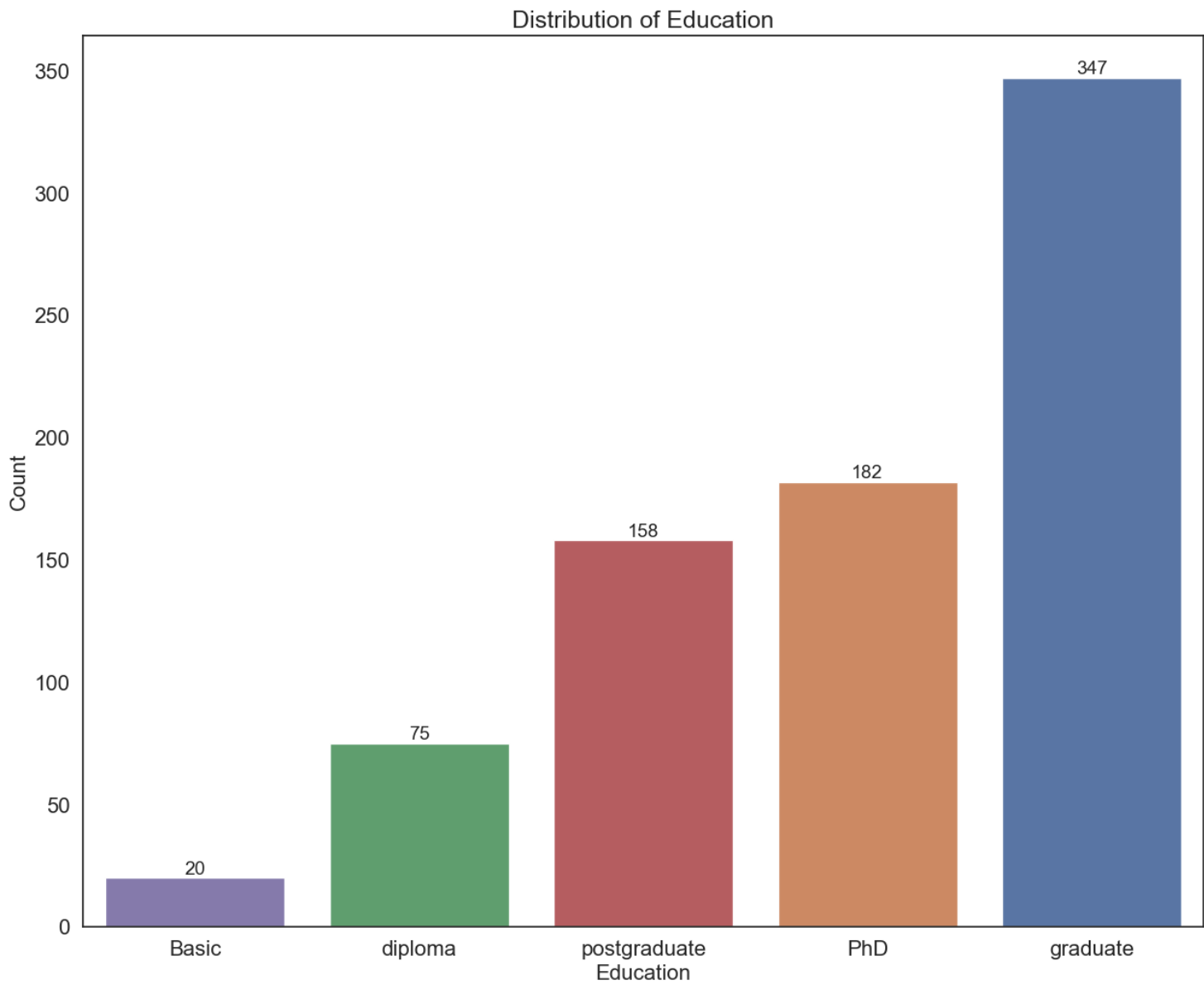
Boxplot of loyalty points by gender

Distributions very similar between genders: both have high outliers



Education

- 87% (687) of 782 customers attended university and 97% (762) completed their school education.

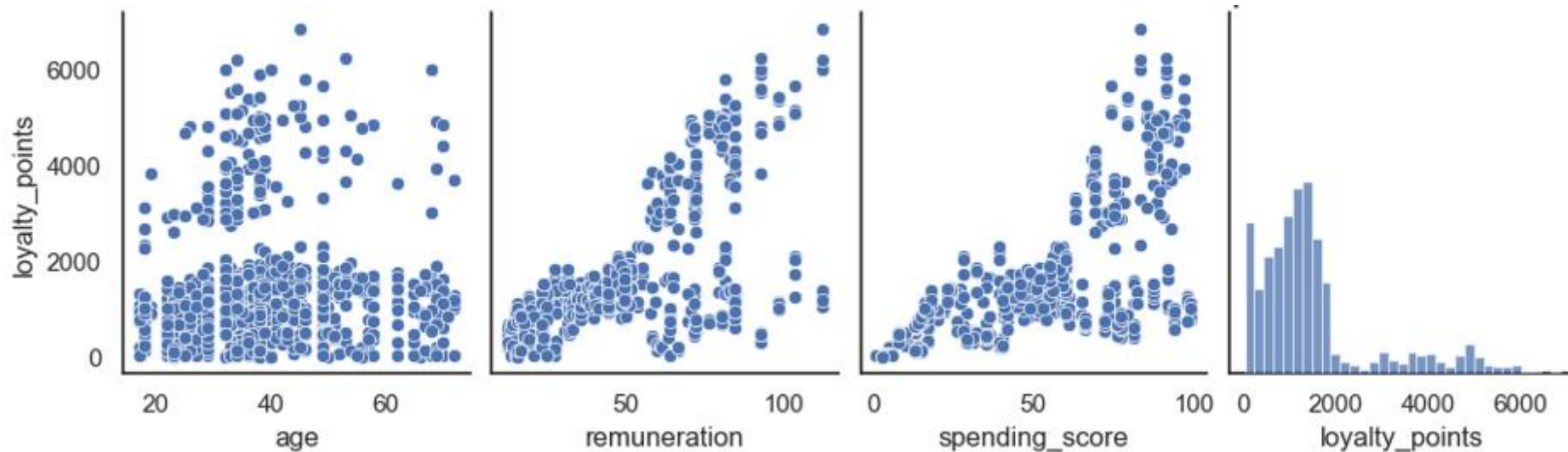


Visualisations and insights

Linear regression with loyalty points as the dependent variable

- Pairplot suggests spending score and remuneration have a linear relationship with loyalty
- Correlation matrix shows age has a weaker influence.

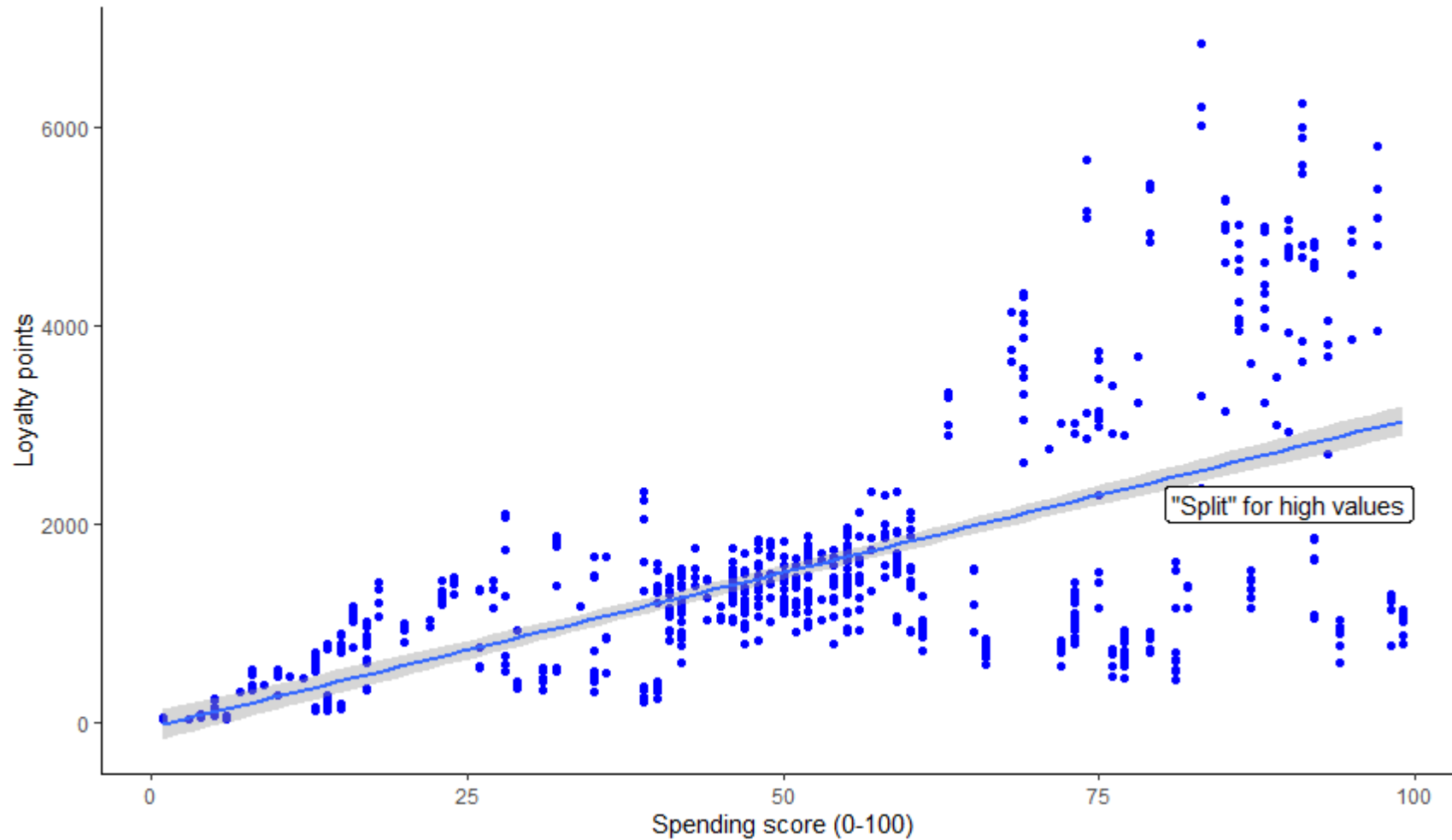
	age	remuneration	spending_score	loyalty_points
age	1.000000	0.006024	-0.124630	0.028904
remuneration	0.006024	1.000000	-0.001645	0.630901
spending_score	-0.124630	-0.001645	1.000000	0.632116
loyalty_points	0.028904	0.630901	0.632116	1.000000



- Spending score only as an independent variable showed R-squared 40% and MAE 730. “Splitting” of the data into higher and lower loyalty clusters with higher spending noted.

Linear regression using spending score to predict loyalty points

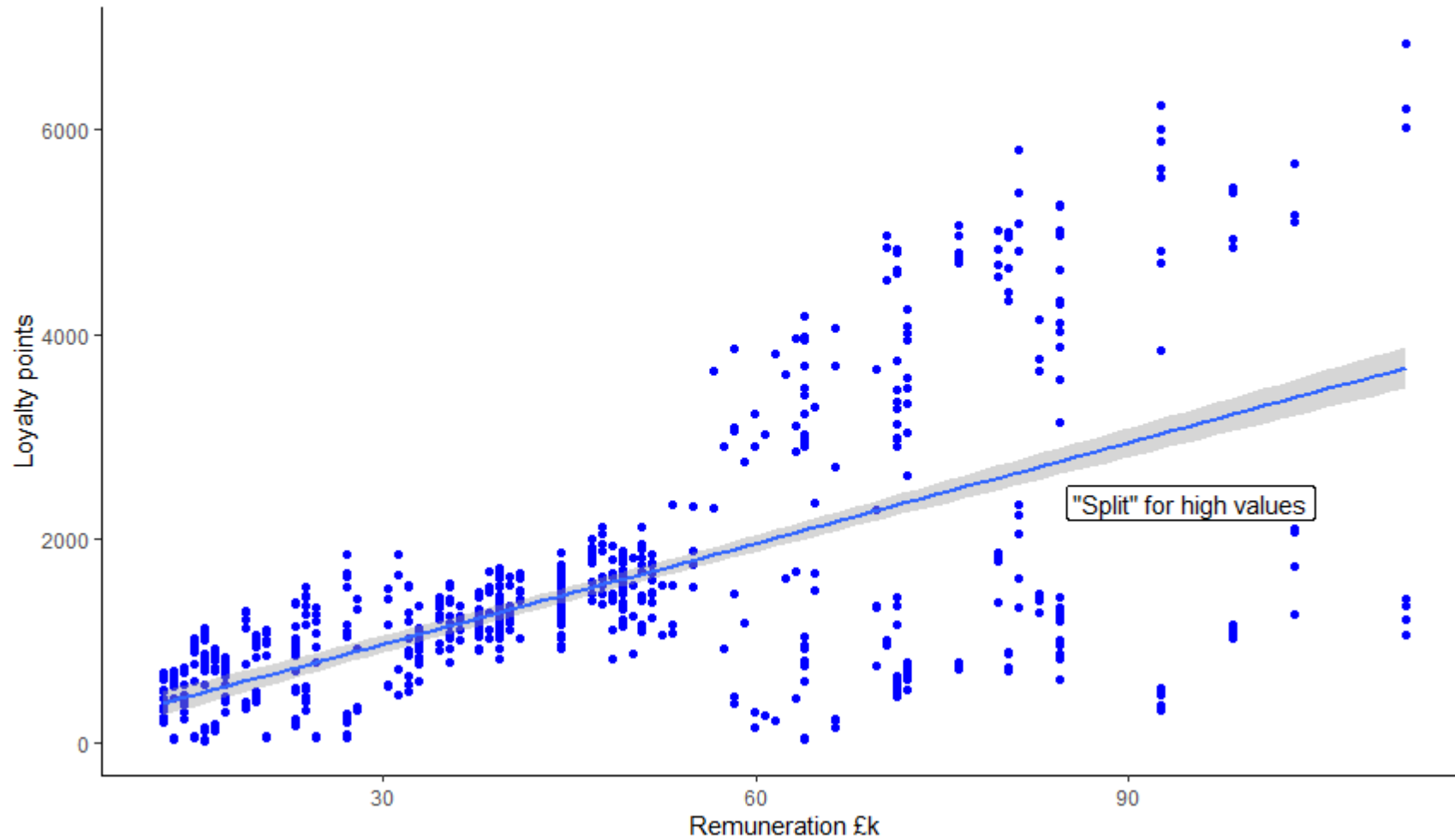
Appears to show a reasonable correlation at low values and then "split" for high values. Shaded area shows 95% conf. int



- Remuneration as the independent variable showed R-squared 39% and MAE 717, again underfitting the data.

Scatterplot of loyalty points and remuneration

Shows a reasonable correlation at low values and then "split" for high values. Shaded area shows 95% conf. int



- Age showed an R-squared of 0.2% so it was not considered further.

An MLR model using both variables showed R-squared 80% and MAE 485 - against the mean (1497), this error is high.

The best MLR model was a log-log transform:

$$\log(\text{loyalty points}) = -0.4655 + 0.99(\log(\text{spending score})) + 1.0186(\log(\text{remuneration}))$$

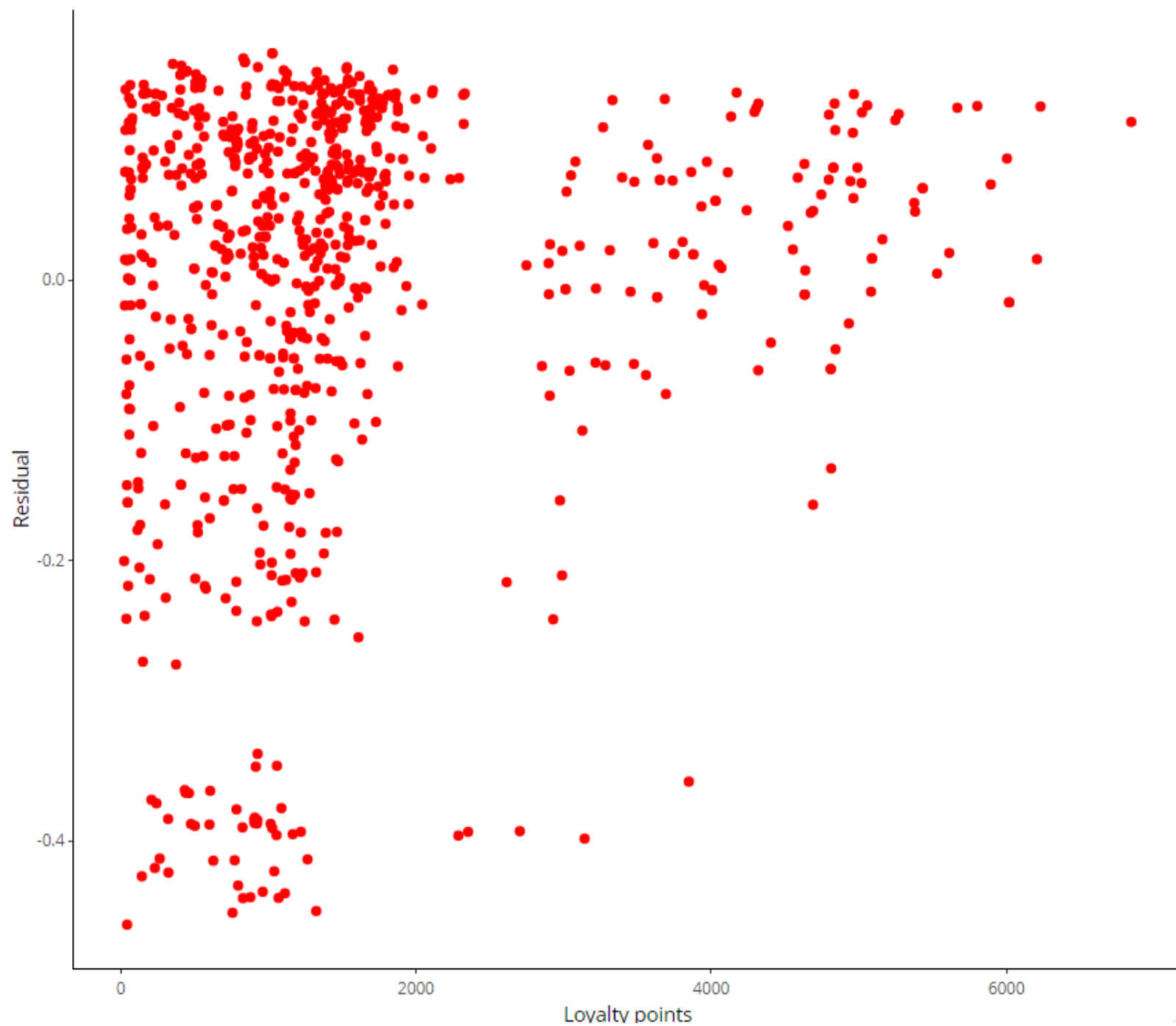
R-squared of 98%, MAE 167, RMSE-MAE 112 shows that larger errors remain despite the lower MAE.

- Assumptions of linear regression met except for residual normality.
- Unless the model is used for hypothesis testing, this is not a concern (ref. 5).
- VIF test showed no multicollinearity
- Breusch-Pagan test showed homoscedasticity
- Durbin-Watson test showed negligible autocorrelation.

Because the coefficients are ≈ 1 the model can be approximated as: (see ref. 6)

A 1% increase in spending score AND a 1% increase in remuneration is associated with an average increase of 1% in loyalty points.

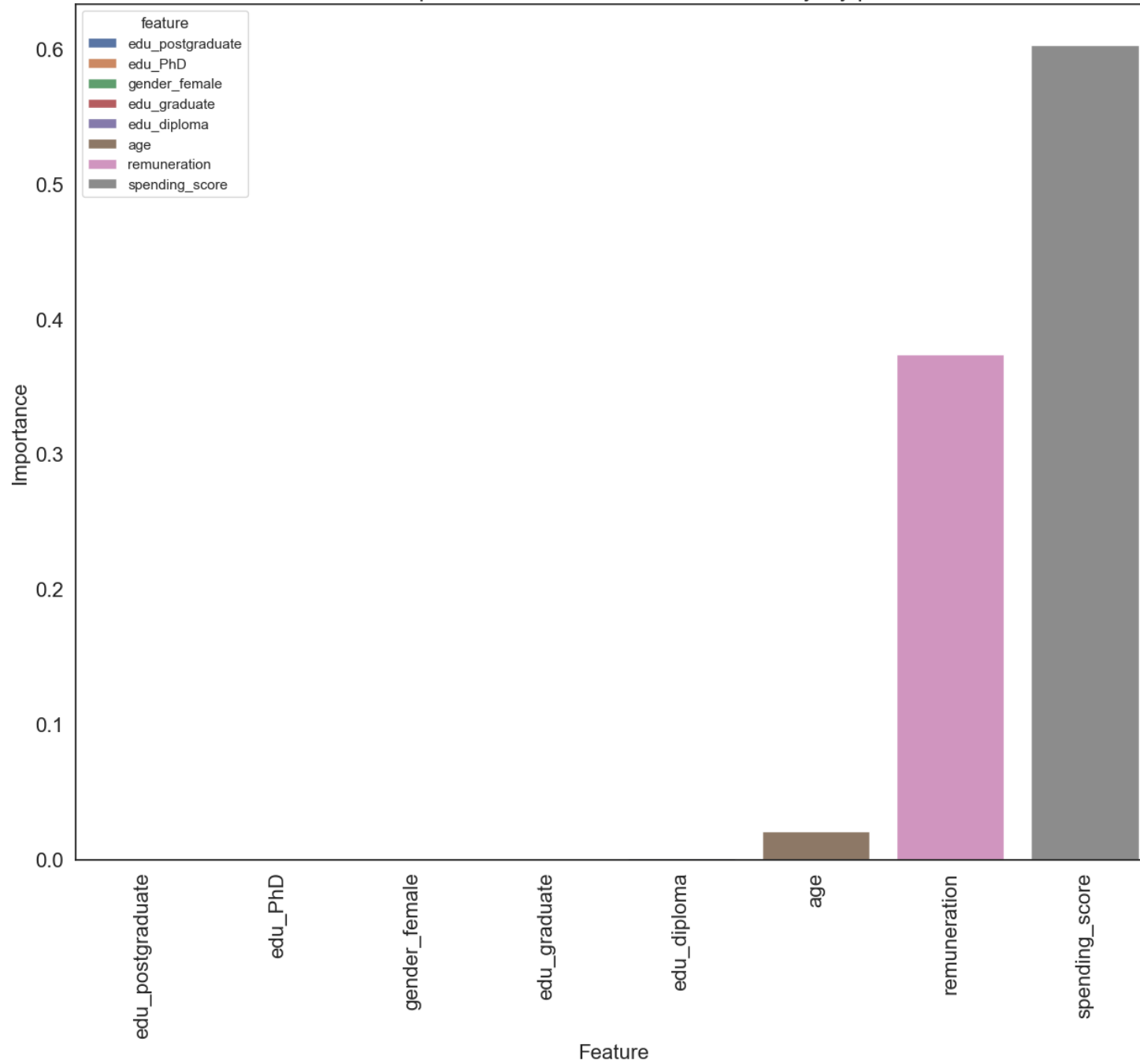
Distribution of residuals in log-log model by loyalty points: no heteroscedasticity



Decision Tree Regression (DTR) with loyalty points as the dependent variable

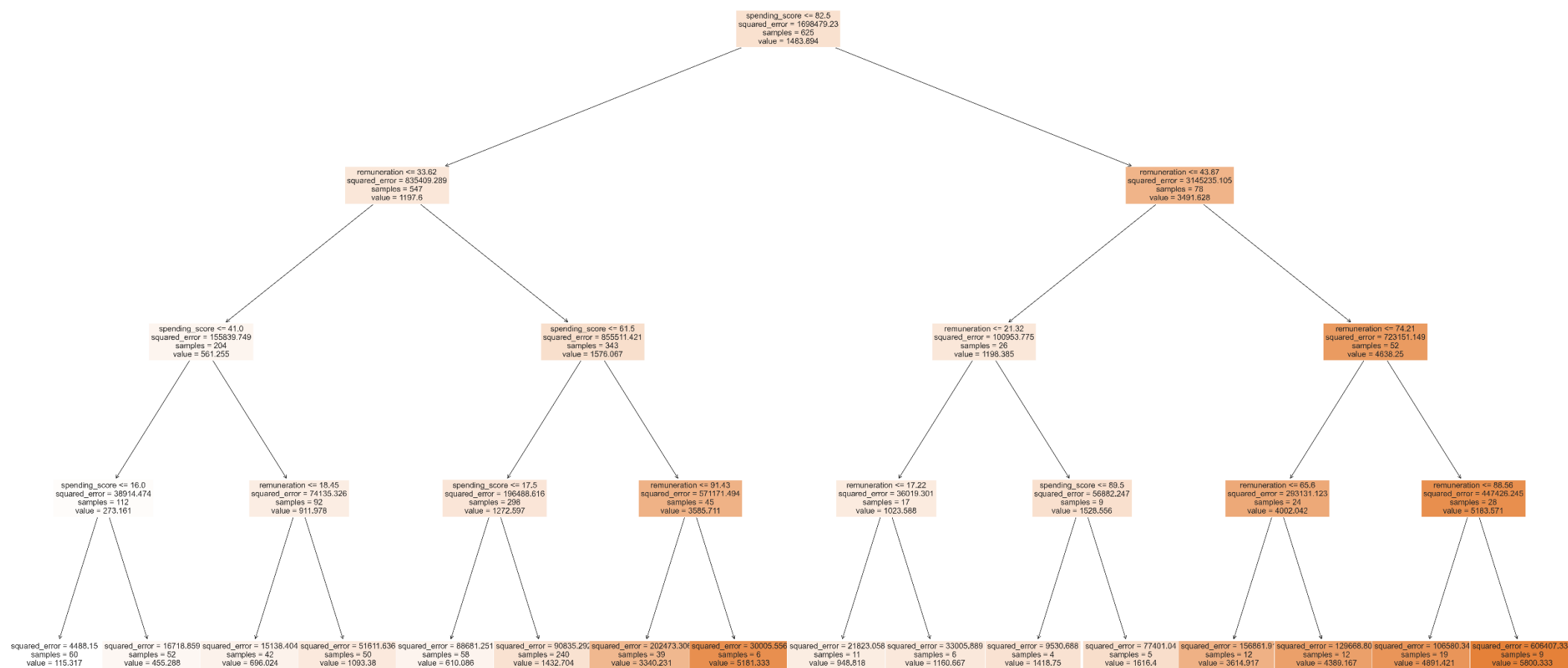
- Education and Gender “one-hot” encoded to convert categorical values into Boolean fields, removing one value to eliminate multicollinearity.
- Unpruned tree used to determine feature importances.
- Only spending score and remuneration had appreciable importance - all others were removed.

Feature importances in decision tree model of loyalty points



Hyperparameter tuning identified optimal values of maximum depth (12), minimum leaf sample size (3), and node split sample size (7). Maximum depth was reduced to 4 to trade off accuracy and complexity (R squared 93%, MAE 251, RMS-MAE 126). The final tree is shown below.

Decision Tree: remuneration and spending score only, max_depth=4

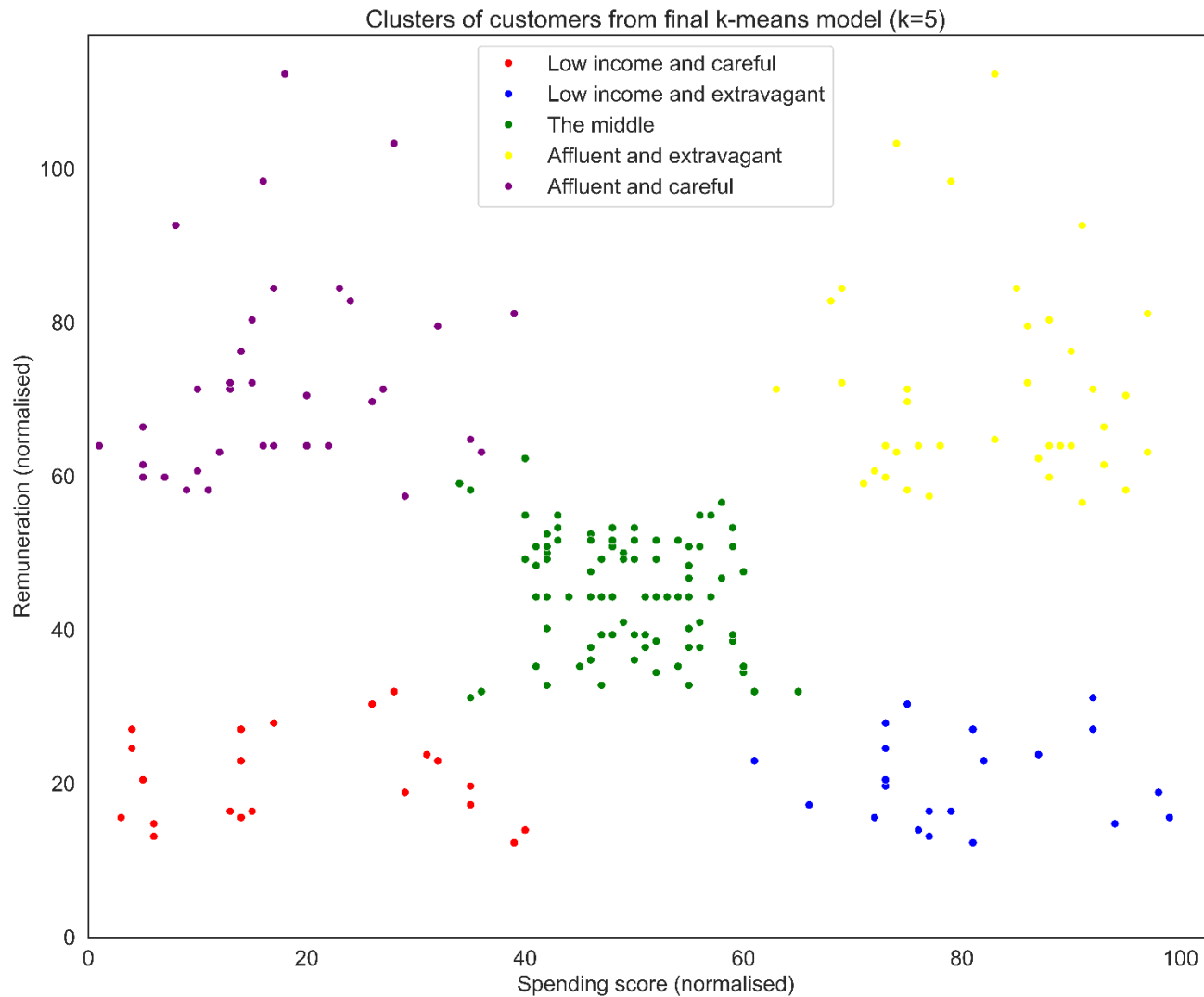


- Random Forest Regression (RFR) was investigated
- RFR had marginal impact on the R-squared (96%). MAE 159, RMS-MAE 120.
- RFR is an aggregation of trees, so not as interpretable and can only be used to predict inside an application or code notebook.

- No real benefit in losing the interpretability of DTR for a small accuracy gain

Cluster analysis of spending score and remuneration

- K-means clustering using elbow and silhouette methods, and hierarchical clustering suggests 5 as the optimal number of clusters
- These can serve as “buyer personas”

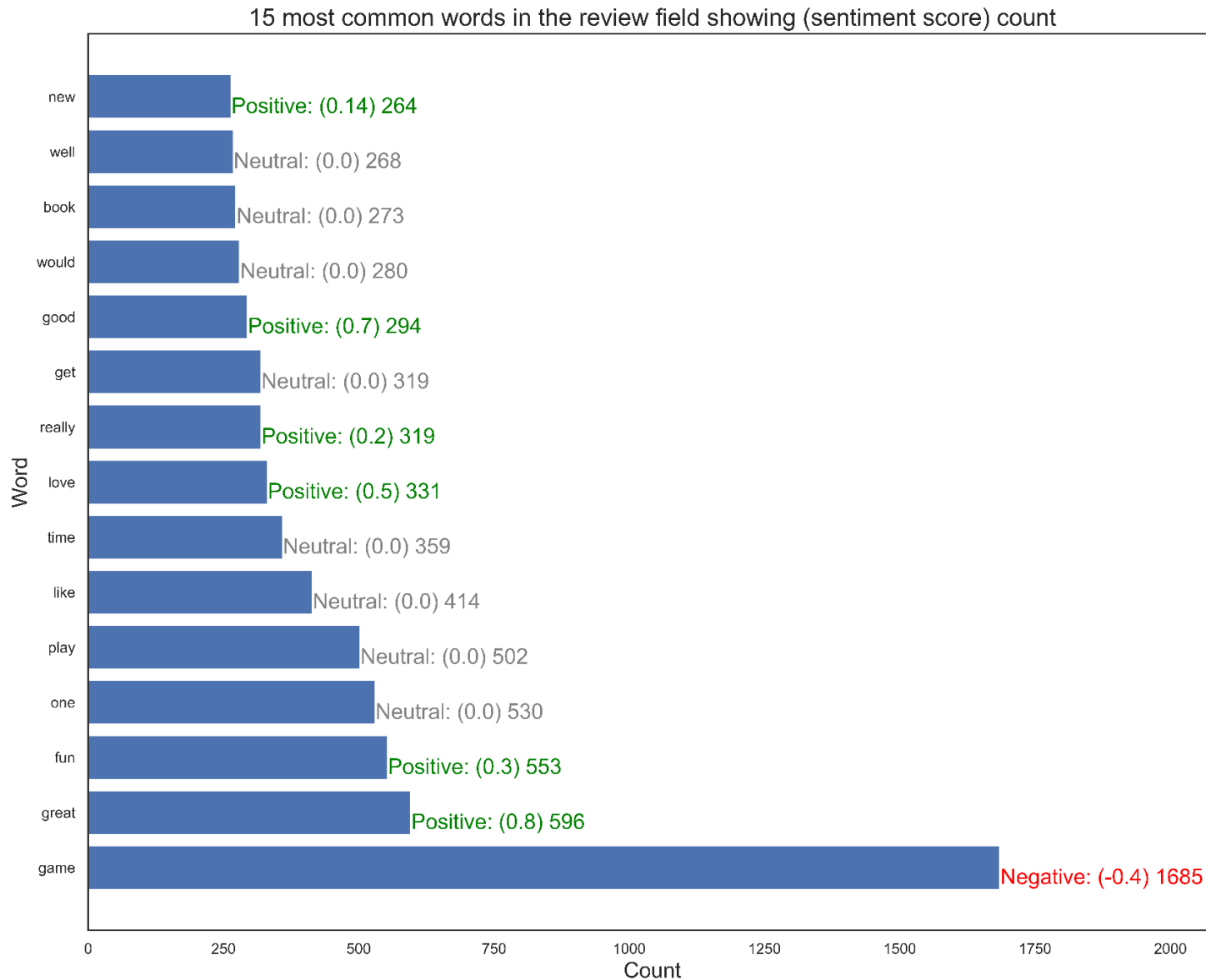


Review sentiment analysis using NLP techniques

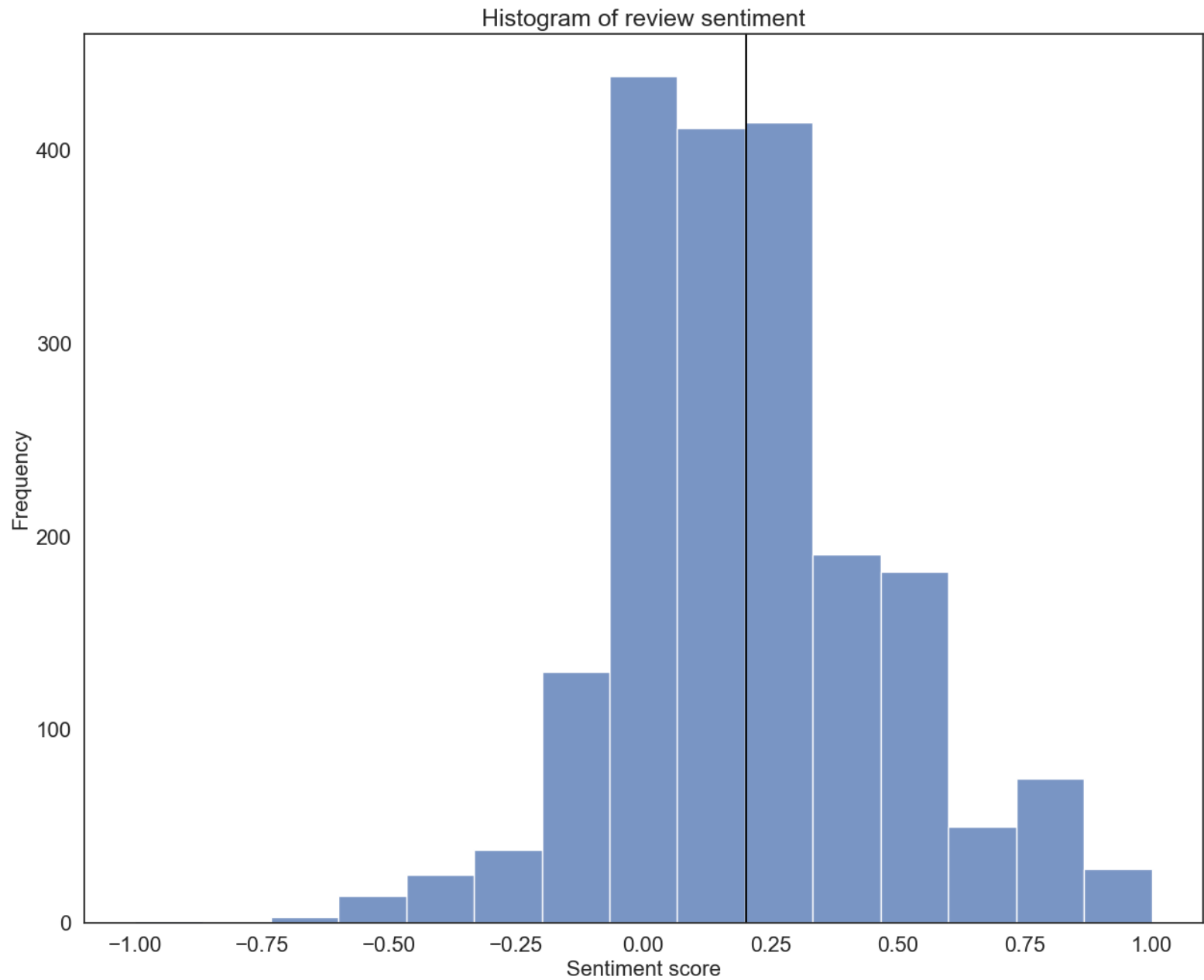
- Reviews and summaries lowercased and punctuation/stopwords removed, then tokenized and lemmatized
- Word cloud of the review tokens is shown below



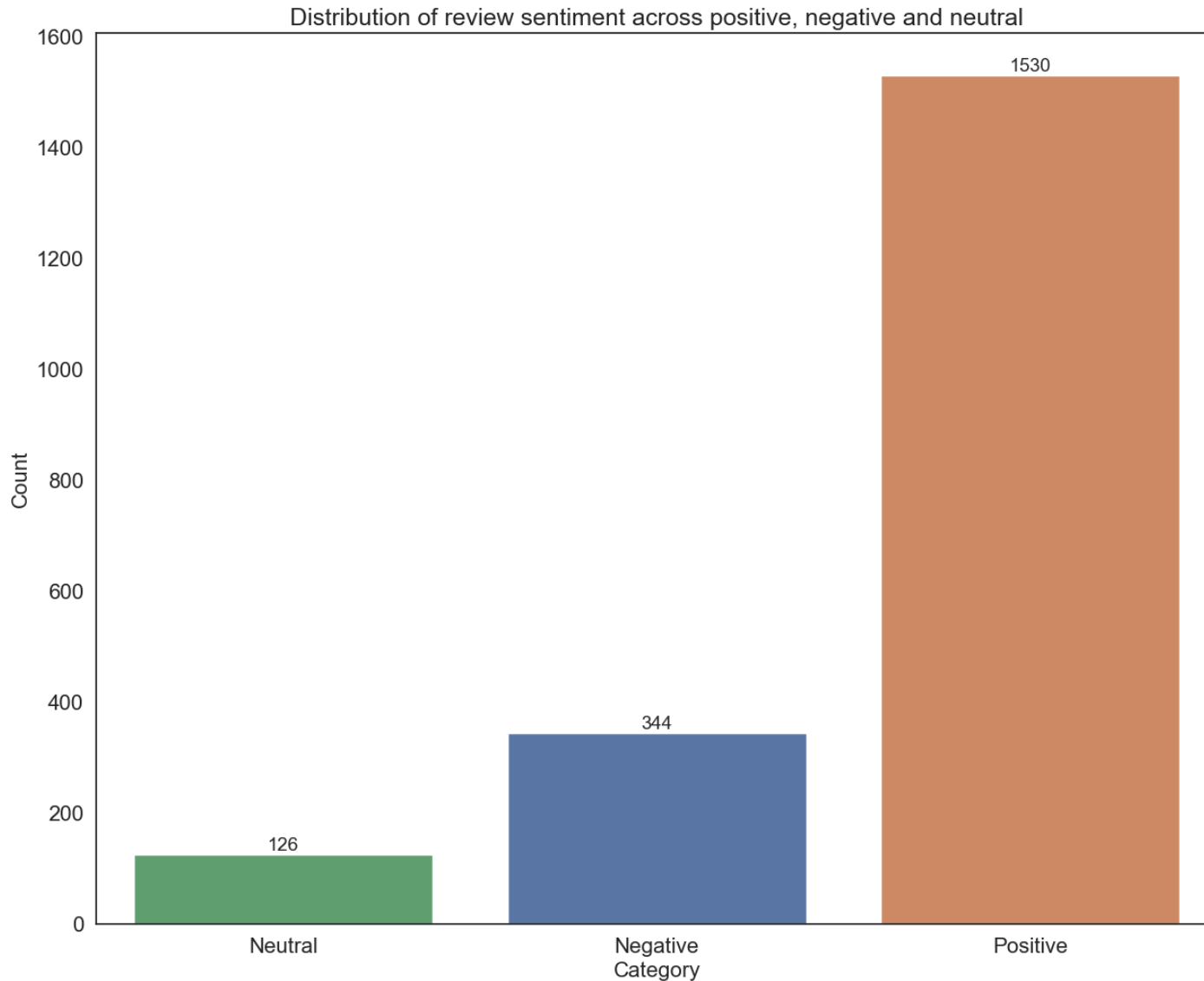
- 10 most common words from reviews shown below.
- Largely positive (sentiment score>0). “Game” interpreted as negative (score<0), possibly incorrectly.
- Summaries were similar.



A histogram of the review sentiment shows the overall score to be mildly positive (mean polarity 0.2). The summaries were similar.



- Top 20 positive and bottom 20 negative reviews were extracted and are available on request.
- Using a simple “bag-of-words” approach “game”, “great” and “fun” were the three most common words in a document term matrix. A similar approach limited to bigrams showed “year old”, “great game” and “fun game” as the top three.
- Counts of positive, negative and neutral reviews showed positive sentiment far outweighs negative.



Analysis of products with the highest proportion of negative reviews (discounting neutrals) gave the below results.

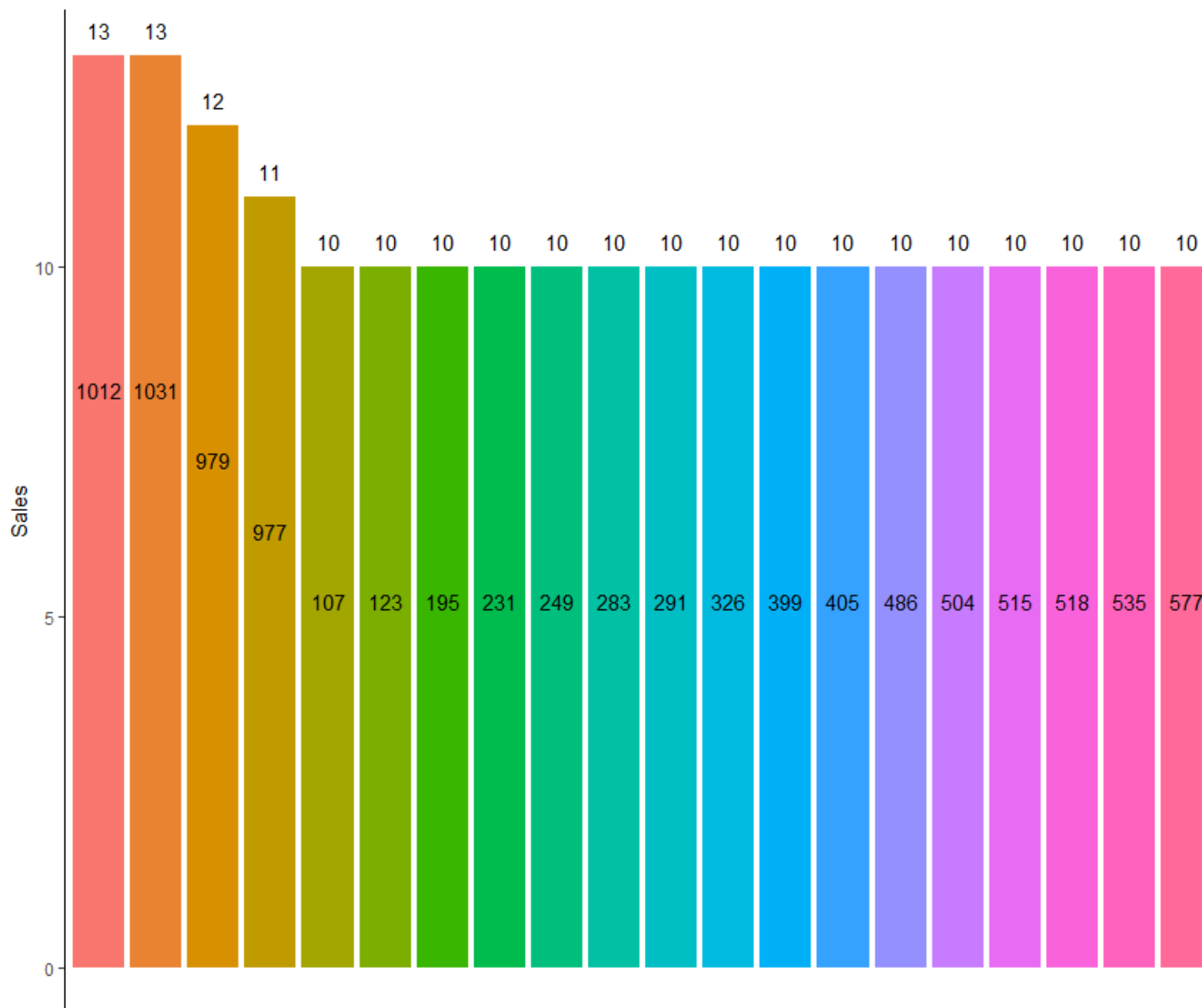
Negative review percentage	
Product Id	
2795	70.000000
4399	66.670000
9597	55.560000
466	50.000000
2324	50.000000
2387	50.000000
5512	50.000000
1945	44.440000
2870	44.440000
3896	44.440000

Analysis of the top selling products

A simple bar chart of the best selling product ids is shown below.

Sales of top 20 best selling product ids

Product ids shown inside bars



Insights and recommendations

- Initial data exploration suggests that a typical Turtle Games customer has a university education, is early middle-aged, has an above-average salary, and is slightly more likely to be female.

Factors influencing loyalty points and their prediction

- DTR and MLR models show that as customers increase their spending score and remuneration together, they accumulate more loyalty points.***
- DTR less accurate than MLR (R squared 92% vs 98%, MAE 251 vs 167 , RMSE - MAE 126 vs 111).
- RFR slightly more accurate, but cannot be implemented using simple logic in e.g. a spreadsheet as MLR could be
- We recommend that the log-log linear model is employed to predict loyalty points due to its accuracy and ease of implementation.***

Buyer personas

- Cluster analysis shows 5 “buyer personas”. Suggested marketing strategies are shown below.***

Buyer persona	Suggested marketing strategy
Low income and careful	Introduce an “essentials”, “classic” or “no-frills” range at lower prices
Low income and extravagant	Introduce “easy payment terms”, “buy now pay later” <u>etc</u> on premium product lines
The middle	“Silver club” offering a cheaper version of the Platinum club (see below)
Affluent and extravagant	Introduce a Gold range of premium products. Leather-bound cases, aged faux-ivory dice, embossed boards etc. Introduce a Platinum club with a monthly membership fee. Early access to new releases, special editions, Q and A events with game designers....
Affluent and careful	Introduce “discount days” for the premium product lines to tempt the group into indulging themselves.

Review sentiment analysis

- Overall sentiment was mildly positive (mean score 0.2)
- No clearly negative words were in the top 15 most common
- 77% of reviews were positive
- ***Deeper analysis of the 10 products with the highest proportion of negative reviews may be helpful to Turtle Games and lead to opportunities to minimise future negative reviews that can damage the brand***

Best-selling products

- ***“Special editions” of the popular products shown in the bar chart may drive enthusiasts to make repeat purchases if the edition has high quality bonus content not available originally.***

Limitations

- No data on product type – board game, console etc
- Variables outside the dataset may influence loyalty points or personas
- NLP may no longer be an optimal approach for sentiment analysis in future

Future ideas

- Analysis of product type sales to different demographics
- LLM's for sentiment analysis
- Refine models using any additional customer data Turtle Games has

References

1. “Crafting [buyer or user personas](#) is fundamental for successful B2C product marketing. Personas are fictional representations of your ideal customers, and they’re created through extensive research into the needs, behaviours, motivations, and goals of your users.”..
Product Marketing Alliance
<https://www.productmarketingalliance.com/b2c-product-marketing-strategies-boost-growth/>
2. “[Online reviews](#), which significantly influence product sales...”
Exploring the influence of online reviews and motivating factors on sales: A meta-analytic study and the moderating role of product category
Kunlin Li, Yuhan Chen, Liyi Zhang

<https://www.sciencedirect.com/science/article/abs/pii/S0969698919304011>

3. “Creating a cohesive, holistic value equation across pricing and loyalty levers can be the key to propel the next horizon of growth for customer-facing businesses”
Delivering greater values through loyalty and pricing, McKinsey
<https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/members-only-delivering-greater-value-through-loyalty-and-pricing>
4. Average UK salary of £37430 in 2025.
<https://www.forbes.com/uk/advisor/business/average-uk-salary-by-age/>
5. Normally distributed residuals is not mandatory in a linear regression model unless the model is to be used for hypothesis testing. See item 7 on <https://statisticsbyjim.com/regression/ols-linear-regression-assumptions/>
6. Mathematical basis for interpreting the log-log linear regression model as “**If there is a 1% increase in spending score AND a 1% increase in remuneration then that is associated with an approximate average increase of 1% in loyalty points.**”
See <https://quantifyinghealth.com/interpret-log-transformations-in-linear-regression/>

Abbreviations

MAE – mean absolute error

RMSE – square root of the mean squared error

MLR – multiple linear regression

DTR – decision tree regression

RFR – random forest regression

VIF – variance inflation factor

